

BASE CHANGE FOR COCHAINS ON FINITE-TYPE SPACES

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ABSTRACT. We prove that cochains on a CW complex with finitely many cells in each dimension commute with extension of scalars between bounded-above E_∞ -rings. Neither connectivity nor finite relative Tor amplitude is assumed. The proof uses uniform presentation bounds over the Steenrod algebra to establish nilpotence for sequences of spectra with finite p -primary homotopy in intervals of fixed width. Perfect approximations then give a comparison over sphere Postnikov sections, from which the theorem follows by bounded-above dévissage. Two rational examples show that neither upper bound can be omitted: the comparison need not be surjective or injective on π_0 , even when the other coefficient ring is $H\mathbb{Q}$. For fixed A and X , we also prove that comparison with every commutative A -algebra is equivalent to perfectness of the A -module of chains on X . Without any upper bounds, the comparison remains valid after taking the inverse limit of the finite-skeleton comparisons.

1. INTRODUCTION

For a space X and a spectrum M , let

$$C^*(X, M) = F(\Sigma_+^\infty X, M),$$

where F is the function spectrum. A map of commutative ring spectra $A \rightarrow B$ induces a canonical comparison

$$C^*(X, A) \otimes_A B \longrightarrow C^*(X, B). \quad (1.1)$$

All tensor products are derived. When X is a finite CW complex, duality makes (1.1) an equivalence for arbitrary coefficients. For an infinite CW complex, cochains involve an inverse limit over finite subcomplexes, and extension of scalars need not preserve that limit. We prove the following result.

A CW complex is of *finite type* if it has finitely many cells in each dimension. A spectrum is *bounded above* if its homotopy groups vanish in all sufficiently large degrees; no restriction is placed on negative homotopy groups.

Theorem 1.1. *Let X be a finite-type CW complex, and let $A \rightarrow B$ be a map of E_∞ -rings whose underlying spectra are bounded above. Then (1.1) is an equivalence. In particular, this holds for truncated connective E_∞ -rings.*

The connective case answers Question 12 of Antieau [Ant]. The theorem requires neither connectivity of A and B nor finite relative Tor amplitude. The comparison is a map of commutative B -algebras, and we prove that it is an equivalence by checking its underlying spectrum.

The upper bounds on the two coefficient rings cannot be dropped independently. In Section 6, we use the single finite-type space $\bigvee_{n \geq 1} S^{2n}$ to give two counterexamples with connective rational coefficients. In the first example, the source ring is $H\mathbb{Q}$ and the target is unbounded above; the comparison on π_0 is the inclusion $\bigoplus_{n \geq 0} \mathbb{Q} \hookrightarrow \prod_{n \geq 0} \mathbb{Q}$. In the second example, the target is $H\mathbb{Q}$ and the source is unbounded above; the kernel of the comparison contains $\prod_{n \geq 1} \mathbb{Q} / \bigoplus_{n \geq 1} \mathbb{Q}$. Both examples have finite-dimensional homotopy groups in each degree.

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There is also an exact criterion when the target algebra is unrestricted. Theorem 2.2 shows that, for fixed A and X , comparison holds for every commutative A -algebra if and only if $C_*(X; A) = A \otimes_{\Sigma_+^\infty} X$ is a perfect A -module. This recovers the finite-CW case and separates the unrestricted coefficient problem from Theorem 1.1. Proposition 7.1 identifies the inverse limit of the finite-skeleton comparisons with $C^*(X, B)$ for every map $A \rightarrow B$.

Proof of the bounded-above comparison: the reductions. Write $\mathbb{S}$ for the sphere spectrum and $\tau_{\leq n}$ for Postnikov truncation. The first comparison to establish is

$$F(P, \tau_{\leq n}\mathbb{S}) \otimes_{\tau_{\leq n}\mathbb{S}} H\mathbb{F}_p \longrightarrow F(P, H\mathbb{F}_p)$$

for every spectrum P , integer $n \geq 0$, and prime p . We construct a sequence beginning at $H\mathbb{F}_p$ whose successive cofibers are perfect over $\tau_{\leq n}\mathbb{S}$, and whose homotopy groups are finite p -groups in intervals of fixed width tending to infinity. Proposition 3.1 shows that every stage maps trivially to some later stage as a spectrum. Its proof first bounds the presentation degrees of certain right Steenrod modules, then uses a finite sub-Hopf algebra to make blocks of maps into ghosts. A finite-extension argument makes sufficiently long composites of these blocks zero.

The comparison defect is unchanged along the sequence, while both its tensor and function-spectrum terms have zero colimit. This proves the displayed comparison. The finite Postnikov tower of the fiber of $\tau_{\leq n}\mathbb{S} \rightarrow H\mathbb{Z}$ then gives the comparison with $H\mathbb{Z}$. For $P = \Sigma_+^\infty X$, finite type makes cochains commute with filtered colimits having a uniform upper homotopy bound. This extends the comparison to every bounded-above $\tau_{\leq n}\mathbb{S}$ -module. Finally, the connective cover of A supplies a factorization $\tau_{\leq n}\mathbb{S} \rightarrow A \rightarrow B$ for a suitable n , and associativity of extension of scalars proves Theorem 1.1.

Conventions. We work in the stable ∞ -category Sp and in stable ∞ -categories of module spectra. All limits and colimits are taken in these ∞ -categories. Brackets $[U, V]$ denote maps in the homotopy category of spectra, and $[U, V]_R$ denotes maps of R -modules. Suspensions are homological:

$$\pi_j(\Sigma^a M) = \pi_{j-a} M.$$

The notation $M \in \mathrm{Sp}_{[a,b]}$ means that $\pi_j M = 0$ for $j \notin [a, b]$. For an abelian group G , write HG for its Eilenberg–Mac Lane spectrum. If R is connective with $\pi_0 R = \mathbb{Z}$, the R -module HG has the action induced by $R \rightarrow H\mathbb{Z}$.

Module categories are stable, and their forgetful functors to spectra are conservative and preserve limits and colimits [Lur, Proposition 7.1.1.4, Corollaries 4.2.3.2 and 4.2.3.7]. For connective R , the usual t -structure on Mod_R is detected on underlying spectra and its heart is the category of $\pi_0 R$ -modules [Lur, Proposition 7.1.1.13]. A *perfect* R -module is an object of $\mathrm{thick}_{\mathrm{Mod}_R}(R)$, the closure of R under finite sums, suspensions, cofibers, and retracts. These are precisely the compact R -modules [Lur, Proposition 7.2.4.2]. In particular, finite spectra are compact. Homotopy groups commute with filtered colimits and detect equivalences.

2. THE COMPARISON AND PERFECTNESS

For a spectrum P , a commutative ring spectrum R , and an R -module M , let

$$\theta_M^R : F(P, R) \otimes_R M \longrightarrow F(P, M) \tag{2.1}$$

be the R -linear map adjoint to evaluation:

$$P \otimes F(P, R) \otimes_R M \longrightarrow R \otimes_R M \simeq M.$$

The function spectrum $F(P, M)$ carries the action induced by that on M . For $P = \Sigma_+^\infty X$ and M a commutative R -algebra, (2.1) is the underlying-module map of (1.1). Define the comparison defect by

$$\Delta_P^R(M) = \mathrm{cofib}(\theta_M^R). \tag{2.2}$$

Lemma 2.1 (Perfect coefficients). *For every P and R , the functor Δ_P^R is exact and vanishes on every perfect R -module. Thus (2.1) is an equivalence for every perfect M , without boundedness assumptions. If an R -linear map $M \rightarrow N$ has perfect cofiber, then $\Delta_P^R(M) \rightarrow \Delta_P^R(N)$ is an equivalence.*

Proof. Tensoring with $F(P, R)$ is exact because it preserves colimits. The functor $F(P, -)$ preserves limits and is exact because the categories are stable. The pointwise cofiber of a natural transformation between exact functors is exact, as finite cofiber constructions commute. Hence Δ_P^R is exact. It vanishes at R , since θ_R^R is the identity under the unit identification. Exactness and preservation of retracts imply that it vanishes on $\text{thick}_{\text{Mod}_R}(R)$. Applying it to a cofiber sequence with perfect third term proves the last assertion. $\square$

For a space X , its A -module of chains is

$$C_*(X; A) = A \otimes \Sigma_+^\infty X.$$

Write F_A for the internal function object in Mod_A .

Theorem 2.2 (Comparison for all target algebras). *Fix a commutative ring spectrum A and a space X , with no boundedness or finiteness assumptions. The following are equivalent:*

- (i) *For every commutative A -algebra B , the canonical map $C^*(X, A) \otimes_A B \rightarrow C^*(X, B)$ is an equivalence.*
- (ii) *For every A -module N , the canonical map $C^*(X, A) \otimes_A N \rightarrow C^*(X, N)$ is an equivalence.*
- (iii) *The A -module $C_*(X; A)$ is perfect.*

Equivalently, (iii) says that $C_(X; A)$ is dualizable as an A -module.*

Proof. The tensor–function adjunction identifies

$$F_A(C_*(X; A), N) \simeq C^*(X, N)$$

as A -modules. Suppose (iii) holds. For fixed N , the class of $Q \in \text{Mod}_A$ for which

$$F_A(Q, A) \otimes_A N \longrightarrow F_A(Q, N)$$

is an equivalence contains A and is closed under finite sums, suspensions, cofibers, and retracts: both sides are exact contravariant functors of Q . It therefore contains every perfect Q , including $C_*(X; A)$. This proves (ii), which immediately implies (i).

Assume (i). For any A -module N , the trivial square-zero extension $A \oplus N$ is a commutative A -algebra with underlying A -module $A \oplus N$. This construction has no boundedness restriction [Lur, Corollary 7.3.4.14 and Remarks 7.3.4.15–7.3.4.17]. Under the finite direct-sum decompositions, naturality identifies the comparison for $A \oplus N$ with

$$\text{id}_{C^*(X, A)} \oplus (C^*(X, A) \otimes_A N \longrightarrow C^*(X, N)).$$

Its second summand is therefore an equivalence, proving (ii).

Finally, (ii) identifies $F_A(C_*(X; A), -)$ with tensor product by $F_A(C_*(X; A), A)$, so it preserves filtered colimits. The underlying-spectrum functor and Ω^∞ also preserve filtered colimits. Consequently $\text{Map}_{\text{Mod}_A}(C_*(X; A), -)$ does as well. Thus $C_*(X; A)$ is compact, hence perfect [Lur, Proposition 7.2.4.2], proving (iii). Perfect modules are exactly the dualizable modules: the tensor description of their function objects gives evaluation and coevaluation, while dualizability gives the same tensor description and hence compactness. This is the duality characterization of [Lur, Proposition 7.2.4.4]. $\square$

Corollary 2.3. *No upper bounds on coefficient rings are needed in either of the following cases:*

- (i) *X is a finite CW complex or is finitely dominated.*
- (ii) *For the specified map $A \rightarrow B$, the underlying A -module B is perfect; in this case X can be any space.*

Proof. A finite CW structure gives $C_*(X; A)$ a finite cell filtration with subquotients suspensions of A . A finite domination makes it a retract of such a module, so (i) follows from Theorem 2.2. Assertion (ii) is Lemma 2.1. $\square$

Corollary 2.4 (Field coefficients). *Let K be any field and $A = HK$. Comparison is an equivalence for every commutative HK -algebra B if and only if*

$$\sum_{d \geq 0} \dim_K H_d(X; K) < \infty.$$

Proof. The homotopy groups of $C_*(X; HK)$ are $H_*(X; K)$. Choose a basis for each group and represent its elements by HK -module maps from suspensions of HK . The resulting map from the corresponding direct sum of suspensions to $C_*(X; HK)$ is an isomorphism on every homotopy group and hence an equivalence. If total homology is finite-dimensional, this sum is finite, so the module is perfect. Conversely, finite sums, suspensions, cofibers, and retracts preserve finite total homotopy dimension for HK -modules. Every perfect HK -module therefore has finite-dimensional total homotopy. Apply Theorem 2.2. $\square$

3. FIXED-WIDTH NILPOTENCE

Throughout this section, fix a prime p and put $k = H\mathbb{F}_p$.

Proposition 3.1 (Fixed-width nilpotence). *Fix p and $w \geq 0$. Let*

$$M_0 \xrightarrow{f_0} M_1 \xrightarrow{f_1} M_2 \longrightarrow \cdots \quad (3.1)$$

be a sequence of spectra such that $M_i \in \mathrm{Sp}_{[i, i+w]}$ and all its homotopy groups are finite abelian p -groups. For every i , there is $j > i$ for which the composite $M_i \rightarrow M_j$ is nullhomotopic as a map of spectra.

The proof first bounds the presentation degrees of the right Steenrod modules of maps from k , uniformly for spectra of fixed homotopy width. The resulting generator-degree bounds put the matrices of all transition maps into a single finite-dimensional sub-Hopf algebra. A degree estimate then shows that a fixed number of consecutive maps induces zero on maps from every shift of k . A finite filtration of each source by shifts of k turns sufficiently many of these blocks into a nullhomotopic composite. The block length depends only on p, w ; the number of blocks may depend on the source.

3.1. Uniform presentation bounds over the Steenrod algebra. The graded algebra

$$\mathcal{A}^q = [k, \Sigma^q k]$$

is the mod- p Steenrod algebra. We multiply operations by composition, with suspensions inserted: if $a \in \mathcal{A}^d$ and $b \in \mathcal{A}^e$, then $ab = (\Sigma^e a) \circ b$.

Proposition 3.2 (Classical structural input). *The algebra $\mathcal{A}$ is nonnegatively graded, with $\mathcal{A}^0 = \mathbb{F}_p$, and each graded piece is finite-dimensional. It is the increasing union of finite-dimensional graded sub-Hopf algebras $\mathcal{A}(m)$. For each m , $\mathcal{A}$ is free, in particular flat, as a left $\mathcal{A}(m)$ -module. Consequently, for every integer $c \geq 0$ there is a finite-dimensional graded sub-Hopf algebra $D \subset \mathcal{A}$ containing $\bigoplus_{0 \leq q \leq c} \mathcal{A}^q$ such that $\mathcal{A}$ is flat as a left D -module.*

Proof. These are the P -algebra properties of the Steenrod algebra. They are recorded, for Steenrod algebras at a prime, in [Bak, Section 1, the definition of a P -Hopf algebra and the paragraphs following it]. The freeness statement follows from [MM, Theorem 4.4]: apply it to $\mathcal{A}(m) \subset \mathcal{A}$ and its quotient $\mathbb{F}_p \otimes_{\mathcal{A}(m)} \mathcal{A}$. The required splittings exist over the field $\mathbb{F}_p$, and the theorem identifies $\mathcal{A}$, as a left $\mathcal{A}(m)$ -module, with $\mathcal{A}(m) \otimes_{\mathbb{F}_p} (\mathbb{F}_p \otimes_{\mathcal{A}(m)} \mathcal{A})$. For the last assertion, the finite-dimensional vector space $\bigoplus_{0 \leq q \leq c} \mathcal{A}^q$ has a finite basis; all of that basis belongs to a single stage of the increasing union. Take that stage for D . $\square$

All modules over $\mathcal{A}$ or D in this section are algebraic graded *right* modules; their tensor products are ordinary algebraic tensor products, not tensor products of spectra. For a graded module V , define a shift by

$$(V\langle s \rangle)^q = V^{q-s}.$$

Thus a generator of degree d in V has degree $d + s$ in $V\langle s \rangle$.

Definition 3.3. For integers $L \leq U$, a graded $\mathcal{A}$ -module has an $[L, U]$ -presentation if it admits a presentation

$$\bigoplus_{\nu=1}^a \mathcal{A}\langle r_\nu \rangle \longrightarrow \bigoplus_{\mu=1}^b \mathcal{A}\langle g_\mu \rangle \longrightarrow V \longrightarrow 0, \quad (3.2)$$

where a, b are finite, the map has degree zero, and all r_ν, g_μ belong to $[L, U]$. Empty sums are allowed. There is no requirement that the first map be injective.

Lemma 3.4 (Bounded presentation calculus). *Fix $L \leq U$.*

- (i) *There is an integer $K = K(p, L, U) \geq U$ such that, for every degree-zero map $f : V \rightarrow W$ between modules with $[L, U]$ -presentations, both $\ker f$ and $\operatorname{coker} f$ have $[L, K]$ -presentations. The same K works independently of the numbers of generators and relations and of the map.*
- (ii) *If $0 \rightarrow V \rightarrow E \rightarrow W \rightarrow 0$ is exact and V, W have $[L, U]$ -presentations, then E has an $[L, U]$ -presentation.*

Finite direct sums preserve presentation bounds, and shifting by s changes $[L, U]$ to $[L + s, U + s]$.

Proof. For (i), choose, using Proposition 3.2, a finite-dimensional sub-Hopf algebra $D \subset \mathcal{A}$ containing all operations of degrees at most $U - L$. Write

$$d = \max\{q : D^q \neq 0\}.$$

Choose presentations $F_1^V \xrightarrow{d_V} F_0^V \rightarrow V \rightarrow 0$ and $F_1^W \xrightarrow{d_W} F_0^W \rightarrow W \rightarrow 0$ as in (3.2). Since the F_i are graded free modules, lift f to a degree-zero map

$$\tilde{f}_0 : F_0^V \longrightarrow F_0^W.$$

The image of $\tilde{f}_0 d_V$ lies in $\operatorname{im} d_W$. Projectivity of F_1^V therefore gives a degree-zero lift $\tilde{f}_1 : F_1^V \rightarrow F_1^W$ with

$$d_W \tilde{f}_1 = \tilde{f}_0 d_V. \quad (3.3)$$

Every nonzero matrix entry of these four maps has degree equal to the degree of a source generator minus that of a target generator. Its degree is consequently between 0 and $U - L$, so all entries lie in D . Equation (3.3) is then an equality of matrices over D , since $D \rightarrow \mathcal{A}$ is injective.

The presentations and the lifted maps thus define finite-dimensional graded D -modules V_D, W_D and a map $f_D : V_D \rightarrow W_D$ such that

$$(V, W, f) \cong (V_D \otimes_D \mathcal{A}, W_D \otimes_D \mathcal{A}, f_D \otimes_D \mathcal{A}).$$

Both V_D and W_D are supported in $[L, U + d]$. Flatness of the left D -module $\mathcal{A}$ gives

$$\begin{aligned} \ker f &\cong (\ker f_D) \otimes_D \mathcal{A}, \\ \operatorname{coker} f &\cong (\operatorname{coker} f_D) \otimes_D \mathcal{A}. \end{aligned}$$

The D -modules $\ker f_D$ and $\operatorname{coker} f_D$ are finite-dimensional and supported in $[L, U + d]$. For either one, take a homogeneous vector-space basis as a set of module generators. The resulting finite free D -module is supported in $[L, U + 2d]$. Its kernel is finite-dimensional in the same interval, and a homogeneous basis of that kernel is a finite set of relation generators. This gives a D -presentation whose generator and relation degrees belong to $[L, U + 2d]$. Extension of scalars gives the desired $\mathcal{A}$ -presentation. Thus one may take

$$K(p, L, U) = U + 2d. \quad (3.4)$$

The choice of D , hence of d , depends only on p, L, U .

For (ii), choose the indicated presentations of V and W , and lift the generators of W homogeneously to E . Together with the generators of V , these give a finite generating set for E in degrees $[L, U]$. Each chosen relation of W , evaluated on its lifts, lies in V ; express it in the chosen generators of V , in its original homogeneous degree. Subtracting that expression gives a relation of E of the same degree. Use these corrected relations together with the original relations of V . They generate the entire relation module: any relation projects to a relation of W , and subtracting the corresponding corrected relations leaves a relation in the generators of V . All relation degrees are still in $[L, U]$.

The assertions about finite direct sums and shifts follow by performing those operations on the presentations. $\square$

3.2. Maps from the mod- p Eilenberg–Mac Lane spectrum. For a spectrum E , define

$$\mathcal{T}(E)^q = [k, \Sigma^q E]. \quad (3.5)$$

Precomposition makes this a graded right $\mathcal{A}$ -module: for $x : k \rightarrow \Sigma^q E$ and $a : k \rightarrow \Sigma^d k$, put $xa = (\Sigma^d x) \circ a$. These groups are $\mathbb{F}_p$ -vector spaces because $p \text{id}_k$ is nullhomotopic. Indeed, multiplication by p is already zero as a k -module map, and integer multiplication on $[k, \Sigma^q E]$ can be computed on the source. Notice that

$$\mathcal{T}(\Sigma^s E) = \mathcal{T}(E)\langle -s \rangle. \quad (3.6)$$

The functor $\mathcal{T}$ is covariant in E and takes cofiber sequences to long exact sequences of graded right $\mathcal{A}$ -modules, with the indicated shifts. It is not the usual contravariant mod- p cohomology functor.

Lemma 3.5 (A bound independent of the exponent). *For every finite abelian p -group G , the module $\mathcal{T}(HG)$ has an $[0, 2]$ -presentation.*

Proof. Set $J^q = [k, \Sigma^q H\mathbb{Z}]$. Apply $[k, \Sigma^q (-)]$ to

$$H\mathbb{Z} \xrightarrow{p} H\mathbb{Z} \xrightarrow{\rho} k \xrightarrow{\delta} \Sigma H\mathbb{Z}. \quad (3.7)$$

Multiplication by p acts as zero on J^q . We therefore obtain exact sequences

$$0 \longrightarrow J^q \xrightarrow{\rho_*} \mathcal{A}^q \xrightarrow{\delta_*} J^{q+1} \longrightarrow 0 \quad (q \in \mathbb{Z}). \quad (3.8)$$

Let $\beta = (\Sigma\rho)\delta \in \mathcal{A}^1$; this is the Bockstein operation. The composite of the last map in degree q and the first map in degree $q+1$ in (3.8) is left multiplication by β . Since the first map is injective and the last is surjective,

$$\text{im}(\rho_*) = \beta\mathcal{A}, \quad \ker(\beta \cdot - : \mathcal{A} \rightarrow \mathcal{A}\langle -1 \rangle) = \beta\mathcal{A}. \quad (3.9)$$

Thus $J \cong \beta\mathcal{A}$ as graded right modules. In particular, J has a presentation

$$\mathcal{A}\langle 2 \rangle \longrightarrow \mathcal{A}\langle 1 \rangle \longrightarrow J \longrightarrow 0, \quad (3.10)$$

where the first map sends its generator to the generator of $\mathcal{A}\langle 1 \rangle$ multiplied on the right by β , and the second sends its generator to the element corresponding to β . Exactness follows directly from (3.9). In particular, no freeness assertion concerning the Bockstein subalgebra is needed.

For $a \geq 1$, use the cofiber sequence

$$H\mathbb{Z} \xrightarrow{p^a} H\mathbb{Z} \longrightarrow H(\mathbb{Z}/p^a).$$

Since p^a also acts as zero on J , it gives an exact sequence of graded right $\mathcal{A}$ -modules

$$0 \longrightarrow J \longrightarrow \mathcal{T}(H(\mathbb{Z}/p^a)) \longrightarrow J\langle -1 \rangle \longrightarrow 0. \quad (3.11)$$

The two ends have presentations in $[1, 2]$ and $[0, 1]$, respectively. Regard both as $[0, 2]$ -presentations and apply Lemma 3.4(ii). This proves a bound independent of a . Finally, a finite abelian p -group is a finite direct sum of cyclic p -groups. Applying finite direct sums to their presentations proves the assertion. No splitting of (3.11) is asserted. $\square$

Lemma 3.6 (Uniform Postnikov bound). *For every integer $w \geq 0$, there are integers $L_w \leq U_w$, depending only on p, w , such that $\mathcal{T}(E)$ has an $[L_w, U_w]$ -presentation whenever $E \in \mathrm{Sp}_{[0, w]}$ and every $\pi_j E$ is a finite abelian p -group.*

Proof. The case $w = 0$ is Lemma 3.5, so take $L_0 = 0$ and $U_0 = 2$. Suppose bounds L_{w-1}, U_{w-1} have been chosen. Put

$$U = \Sigma^w H(\pi_w E), \quad V = \tau_{\leq w-1} E.$$

The Postnikov cofiber sequence $U \rightarrow E \rightarrow V \rightarrow \Sigma U$ gives

$$0 \longrightarrow \mathrm{coker}(\mathcal{T}(V)\langle 1 \rangle \rightarrow \mathcal{T}(U)) \longrightarrow \mathcal{T}(E) \longrightarrow \ker(\mathcal{T}(V) \rightarrow \mathcal{T}(U)\langle -1 \rangle) \longrightarrow 0. \quad (3.12)$$

All maps inside the kernel and cokernel have degree zero. By (3.6), the presentation bounds for $\mathcal{T}(U)$ are $[-w, 2-w]$. All four modules appearing inside that kernel and cokernel consequently have presentations in the common interval $[L', U']$, where

$$L' = \min\{L_{w-1}, -w-1\}, \quad U' = \max\{U_{w-1} + 1, 2-w\}.$$

Lemma 3.4(i) supplies a common $[L', K(p, L', U')]$ -presentation bound for the kernel and the cokernel. Part (ii) gives the same bound for their extension $\mathcal{T}(E)$. Thus one may define recursively

$$L_w = L', \quad U_w = K(p, L', U').$$

These bounds do not involve the sizes or exponents of the homotopy groups, their numbers of generators, or the Postnikov invariants of E . $\square$

3.3. Ghosts and nilpotence.

Definition 3.7. A map of spectra $f : E \rightarrow E'$ is a k -ghost if $[\Sigma^t k, f] = 0$ for every $t \in \mathbb{Z}$. Equivalently, it is a k -ghost if $\mathcal{T}(f) = 0$.

We give the finite-extension ghost argument explicitly. It is the elementary finite-filtration version of [Chr, Theorem 3.5 and Note 3.6].

Lemma 3.8 (Finite-extension ghost lemma). *Suppose that a spectrum E has a finite filtration whose ℓ successive cofibers are shifts of k . Then any composite of ℓ k -ghosts with source E is nullhomotopic. More generally, the class of spectra from which every a -fold composite of k -ghosts is zero and the analogous class for b -fold composites are closed under extensions with bound $a + b$.*

Proof. For a shift of k , a k -ghost with that source is zero by its definition, applied to the identity of the source. The same assertion holds for a finite sum of shifts of k by restricting to each summand.

For the extension assertion, suppose $U \xrightarrow{u} E \xrightarrow{v} W$ is a cofiber sequence, and that the bounds for U, W are a, b . Consider ghosts

$$E = E_0 \xrightarrow{f_1} E_1 \longrightarrow \cdots \xrightarrow{f_{a+b}} E_{a+b}.$$

The composite of the first a maps, restricted along u , is zero: its first map $f_1 u$ is a ghost, and the remaining maps are ghosts. Exactness of $[-, E_a]$ on the cofiber sequence implies that $f_a \cdots f_1 = h v$ for some $h : W \rightarrow E_a$. Now $f_{a+1} h$ is a ghost. The last b maps, starting with this one, have source W , so their composite is zero. This proves the bound $a + b$. Induction over the filtration proves the first assertion. $\square$

Corollary 3.9. *If E has only finitely many nonzero homotopy groups, all finite abelian p -groups, then there is a finite ghost bound for E . One may use*

$$\ell_p(E) = \sum_j \log_p |\pi_j E|.$$

For $E = 0$ the assertion is understood without any maps.

Proof. Each finite abelian p -group G has a composition series of length $\log_p |G|$ with quotients $\mathbb{F}_p$. Applying H gives a filtration of HG by cofiber sequences with quotients k . The finite Postnikov tower of E assembles these filtrations, with the appropriate shifts. Apply Lemma 3.8. $\square$

Proof of Proposition 3.1. Set $E_i = \Sigma^{-i} M_i$. Lemma 3.6 gives the same presentation bound $[L, U] = [L_w, U_w]$ for all $\mathcal{T}(E_i)$. Choose the resulting finite generating sets. By (3.6), the generators of $\mathcal{T}(M_i)$ have degrees

$$a - i, \quad L \leq a \leq U.$$

Let $F_i \twoheadrightarrow \mathcal{T}(M_i)$ be the graded free module on this finite generating set. Lift $\mathcal{T}(f_i)$ to a degree-zero map $\tilde{f}_i : F_i \rightarrow F_{i+1}$. A nonzero coefficient from a generator of degree $a - i$ to one of degree $b - i - 1$ has degree

$$1 + a - b \in [0, 1 + U - L]. \quad (3.13)$$

Choose a finite-dimensional graded sub-Hopf algebra $D \subset \mathcal{A}$ containing all elements of degrees at most $1 + U - L$, and write $d = \max\{q : D^q \neq 0\}$. The *same* D contains the matrix coefficients of every $\tilde{f}_i$, irrespective of i .

For r consecutive maps, the composite of the lifted matrices has entries in D . Each entry from the generator labelled a at stage i to the generator labelled b at stage $i + r$ is homogeneous of degree

$$r + a - b \geq r + L - U. \quad (3.14)$$

This can also be seen by adding the degrees in (3.13); the intermediate generator degrees cancel. Take

$$r = d + U - L + 1.$$

Every entry in the composite matrix then has degree greater than d , hence is zero. The composite $M_i \rightarrow M_{i+r}$ is therefore a k -ghost, for every i .

If $M_i \neq 0$, let $g_i = \ell_p(M_i)$. The composite $M_i \rightarrow M_{i+rg_i}$ is a composite of g_i of the just-constructed k -ghosts and is nullhomotopic by Corollary 3.9. If $M_i = 0$, any later stage works. The numbers g_i need not be bounded as i varies. $\square$

Corollary 3.10. *Under the hypotheses of Proposition 3.1, for every spectrum P one has*

$$\operatorname{colim}_i M_i \simeq 0, \quad \operatorname{colim}_i F(P, M_i) \simeq 0.$$

Proof. The first assertion follows either from increasing connectivity or from Proposition 3.1. For the second, for every i there is a later j for which the transition map becomes nullhomotopic after applying $F(P, -)$. Hence every element of every homotopy group at stage i dies at a later stage. Homotopy groups commute with the filtered colimit, so that colimit is zero. No commutation of $F(P, -)$ with the original colimit is being asserted. $\square$

4. COMPARISON OVER SPHERE POSTNIKOV SECTIONS

For $n \geq 0$, the sphere Postnikov section $\tau_{\leq n} \mathbb{S}$ is a connective E_∞ -ring with a canonical augmentation to $H\mathbb{Z}$. The existence and the underlying spectrum of this ring Postnikov section follow from [Lur, Propositions 7.1.3.14–7.1.3.15]. We use the classical finiteness theorem that $\pi_j \mathbb{S}$ is finite for every $j > 0$; a reference stating this stable form is [IWX, Chapter 1, p. 1]. In particular,

$$\begin{aligned} \pi_0(\tau_{\leq n} \mathbb{S}) &= \mathbb{Z}, \\ \pi_j(\tau_{\leq n} \mathbb{S}) &\text{ is finite for } 1 \leq j \leq n, \\ \pi_j(\tau_{\leq n} \mathbb{S}) &= 0 \text{ otherwise.} \end{aligned} \quad (4.1)$$

Theorem 4.1 (Mod- p comparison for arbitrary test spectra). *For every $n \geq 0$, every prime p , and every spectrum P , the canonical map*

$$F(P, \tau_{\leq n} \mathbb{S}) \otimes_{\tau_{\leq n} \mathbb{S}} H\mathbb{F}_p \longrightarrow F(P, H\mathbb{F}_p) \quad (4.2)$$

is an equivalence.

Theorem 4.2 (Integral comparison for arbitrary test spectra). *For every $n \geq 0$ and every spectrum P , the canonical map*

$$F(P, \tau_{\leq n}\mathbb{S}) \otimes_{\tau_{\leq n}\mathbb{S}} H\mathbb{Z} \longrightarrow F(P, H\mathbb{Z}) \quad (4.3)$$

is an equivalence.

To prove Theorem 4.1, we construct a sequence beginning at $H\mathbb{F}_p$ whose transition maps have perfect cofibers. By Lemma 2.1, these maps induce equivalences on the comparison cofibers $\Delta_P^{\tau_{\leq n}\mathbb{S}}$. The sequence has zero colimit, so the colimit of the tensor terms vanishes. Its homotopy groups lie in intervals of fixed width tending to positive infinity, so Corollary 3.10 separately makes the colimit of the function terms vanish. This proves the mod- p comparison. The fiber of $\tau_{\leq n}\mathbb{S} \rightarrow H\mathbb{Z}$ is a finite extension of shifts of $H\mathbb{F}_p$ for various primes; exactness then gives Theorem 4.2.

Lemma 4.3. *Fix $n \geq 0$ and a prime p . There is a sequence of $\tau_{\leq n}\mathbb{S}$ -modules*

$$M_0 = H\mathbb{F}_p \longrightarrow M_1 \longrightarrow M_2 \longrightarrow \cdots, \quad (4.4)$$

where $H\mathbb{F}_p$ has its canonical $\tau_{\leq n}\mathbb{S}$ -module structure, such that

- (i) $M_i \in \mathrm{Sp}_{[i, i+n+1]}$ and all its homotopy groups are finite abelian p -groups;
- (ii) $\mathrm{cofib}(M_i \rightarrow M_{i+1})$ is a perfect $\tau_{\leq n}\mathbb{S}$ -module for every i .

Proof. For $e \geq 1$, write

$$(\tau_{\leq n}\mathbb{S})/p^e = \mathrm{cofib}(p^e : \tau_{\leq n}\mathbb{S} \rightarrow \tau_{\leq n}\mathbb{S}).$$

This is a $\tau_{\leq n}\mathbb{S}$ -module; no algebra structure is required. It is perfect and belongs to $\mathrm{Sp}_{[0, n+1]}$. The homotopy exact sequence gives

$$0 \longrightarrow \pi_j((\tau_{\leq n}\mathbb{S})/p^e) \longrightarrow \pi_j((\tau_{\leq n}\mathbb{S})/p^e) \longrightarrow \pi_{j-1}((\tau_{\leq n}\mathbb{S})/p^e) \longrightarrow 0. \quad (4.5)$$

By (4.1), all these groups are finite abelian p -groups.

Start with $M_0 = H\mathbb{F}_p$. Given M_i with property (i), choose $e_i \geq 1$ annihilating $\pi_i M_i$, and choose a finite generating set of that group. Every generator, regarded as an element of

$$[\Sigma^i \tau_{\leq n}\mathbb{S}, M_i]_{\tau_{\leq n}\mathbb{S}} = \pi_i M_i,$$

lifts to a map $\Sigma^i((\tau_{\leq n}\mathbb{S})/p^{e_i}) \rightarrow M_i$. Indeed, applying $[-, M_i]_{\tau_{\leq n}\mathbb{S}}$ to the defining cofiber sequence for $\Sigma^i((\tau_{\leq n}\mathbb{S})/p^{e_i})$ shows that the image of

$$[\Sigma^i((\tau_{\leq n}\mathbb{S})/p^{e_i}), M_i]_{\tau_{\leq n}\mathbb{S}} \longrightarrow [\Sigma^i \tau_{\leq n}\mathbb{S}, M_i]_{\tau_{\leq n}\mathbb{S}}$$

is the kernel of multiplication by p^{e_i} .

The lifts give a map

$$Q_i = \bigoplus_{\text{finitely many}} \Sigma^i((\tau_{\leq n}\mathbb{S})/p^{e_i}) \longrightarrow M_i$$

that is surjective on π_i . If that group is zero, use the empty sum. Define

$$M_{i+1} = \mathrm{cofib}(Q_i \rightarrow M_i). \quad (4.6)$$

The homotopy exact sequence and surjectivity on π_i show that M_{i+1} is $(i+1)$ -connective. Since both Q_i and M_i are at most $(i+n+1)$ -truncated, their cofiber is at most $(i+n+2)$ -truncated. Its homotopy groups are finite abelian p -groups, since each is an extension of a quotient and a subgroup of such groups. This proves (i) at the next stage. Rotating (4.6) gives

$$\mathrm{cofib}(M_i \rightarrow M_{i+1}) \simeq \Sigma Q_i,$$

which is perfect. Thus the induction also establishes (ii). $\square$

Proof of Theorem 4.1. Take the sequence of Lemma 4.3. By Lemma 2.1, every map $\Delta_P^{\tau_{\leq n}\mathbb{S}}(M_i) \rightarrow \Delta_P^{\tau_{\leq n}\mathbb{S}}(M_{i+1})$ is an equivalence. Consequently, the canonical map from the initial term to the colimit is an equivalence:

$$\Delta_P^{\tau_{\leq n}\mathbb{S}}(H\mathbb{F}_p) \simeq \operatorname{colim}_i \Delta_P^{\tau_{\leq n}\mathbb{S}}(M_i). \quad (4.7)$$

This can also be checked on homotopy groups, since all transition homomorphisms are isomorphisms. It remains to show that this colimit vanishes.

The connectivity of M_i tends to infinity, so its colimit is zero on underlying spectra and hence in $\operatorname{Mod}_{\tau_{\leq n}\mathbb{S}}$. Preservation of colimits by relative tensor products therefore gives

$$\begin{aligned} \operatorname{colim}_i (F(P, \tau_{\leq n}\mathbb{S}) \otimes_{\tau_{\leq n}\mathbb{S}} M_i) &\simeq F(P, \tau_{\leq n}\mathbb{S}) \otimes_{\tau_{\leq n}\mathbb{S}} \operatorname{colim}_i M_i \\ &\simeq 0. \end{aligned} \quad (4.8)$$

Separately, the sequence of underlying spectra satisfies Proposition 3.1 with $w = n+1$. Corollary 3.10 gives

$$\operatorname{colim}_i F(P, M_i) \simeq 0. \quad (4.9)$$

Taking the colimit of the cofiber sequences defining $\Delta_P^{\tau_{\leq n}\mathbb{S}}(M_i)$ preserves those cofiber sequences, since cofibers are themselves colimits. Equations (4.7)–(4.9) yield $\Delta_P^{\tau_{\leq n}\mathbb{S}}(H\mathbb{F}_p) = 0$, proving (4.2). $\square$

Remark 4.4. The nullhomotopies supplied by Proposition 3.1 are only asserted in Sp , not in $\operatorname{Mod}_{\tau_{\leq n}\mathbb{S}}$. This suffices for (4.9), which concerns the underlying function spectra. In (4.8), no such nullhomotopy is tensored over $\tau_{\leq n}\mathbb{S}$; one instead uses the actual $\tau_{\leq n}\mathbb{S}$ -module colimit, which is zero on underlying spectra. These are distinct arguments.

Proof of Theorem 4.2. Let $I = \operatorname{fib}(\tau_{\leq n}\mathbb{S} \rightarrow H\mathbb{Z})$ in $\operatorname{Mod}_{\tau_{\leq n}\mathbb{S}}$. It has only finitely many nonzero homotopy groups, all finite abelian groups, in degrees $1, \dots, n$. In the standard t -structure on $\operatorname{Mod}_{\tau_{\leq n}\mathbb{S}}$, its Postnikov layers are the $\tau_{\leq n}\mathbb{S}$ -modules $\Sigma^j H(\pi_j I)$. Each finite abelian group has a finite composition series with quotients $\mathbb{F}_p$ for suitable primes. Applying H and then restricting along $\tau_{\leq n}\mathbb{S} \rightarrow H\mathbb{Z}$ gives cofiber sequences in $\operatorname{Mod}_{\tau_{\leq n}\mathbb{S}}$. Thus I is obtained by finitely many shifts and extensions of the $\tau_{\leq n}\mathbb{S}$ -modules $H\mathbb{F}_p$.

Theorem 4.1 gives $\Delta_P^{\tau_{\leq n}\mathbb{S}}(H\mathbb{F}_p) = 0$ for every prime. By Lemma 2.1, exactness gives $\Delta_P^{\tau_{\leq n}\mathbb{S}}(I) = 0$. Applying the same functor to $I \rightarrow \tau_{\leq n}\mathbb{S} \rightarrow H\mathbb{Z}$ and using $\Delta_P^{\tau_{\leq n}\mathbb{S}}(\tau_{\leq n}\mathbb{S}) = 0$ gives $\Delta_P^{\tau_{\leq n}\mathbb{S}}(H\mathbb{Z}) = 0$, as required. $\square$

5. FINITE-TYPE CONTINUITY AND BOUNDED-ABOVE BASE CHANGE

The finite-type hypothesis enters at this point, not in Theorems 4.1 and 4.2.

Lemma 5.1 (Uniformly bounded-above continuity). *Let X be a finite-type CW complex. If $\{N_\lambda\}$ is a filtered diagram of spectra and there is an integer b such that $N_\lambda \in \operatorname{Sp}_{\leq b}$ for every λ , then*

$$\operatorname{colim}_\lambda C^*(X, N_\lambda) \longrightarrow C^*(X, \operatorname{colim}_\lambda N_\lambda) \quad (5.1)$$

is an equivalence.

Proof. Write $P = \Sigma_+^\infty X$, $P_d = \Sigma_+^\infty X^{(d)}$, and $K_d = \operatorname{cofib}(P_d \rightarrow P)$. The cellular filtration shows that $K_d \in \operatorname{Sp}_{\geq d+1}$: it is built from sphere cells of degrees at least $d+1$, and the connective part of the standard t -structure is closed under extensions and colimits. For $N \in \operatorname{Sp}_{\leq b}$, orthogonality in that t -structure implies

$$F(K_d, N) \in \operatorname{Sp}_{\leq b-d-1}.$$

Indeed its j th homotopy group is $[\Sigma^j K_d, N]$, which vanishes when $j + d + 1 > b$.

Fix $q \in \mathbb{Z}$ and choose $d \geq \max\{0, b - q + 1\}$. The fiber sequence

$$F(K_d, N) \longrightarrow F(P, N) \longrightarrow F(P_d, N)$$

then induces an isomorphism on π_q between its last two terms: both $\pi_q F(K_d, N)$ and $\pi_{q-1} F(K_d, N)$ vanish. The choice of d depends only on b, q .

The spectrum P_d is finite because X is of finite type. Compactness of its suspensions shows that $F(P_d, -)$ preserves filtered colimits on all homotopy groups. The spectrum $\operatorname{colim}_\lambda N_\lambda$ is also at most b -truncated, since homotopy groups commute with filtered colimits. Applying the preceding isomorphism to every N_λ and to their colimit proves that (5.1) is an isomorphism on π_q . This works for every q , proving the lemma. $\square$

Proposition 5.2 (Coefficient dévissage). *Let R be a connective commutative ring spectrum with $\pi_0 R = \mathbb{Z}$. Let X be a finite-type CW complex, put $P = \Sigma_+^\infty X$, and suppose that $\theta_{H\mathbb{Z}}^R$ in (2.1) is an equivalence. Then θ_M^R is an equivalence for every bounded-above R -module M .*

Proof. Call an R -module *good* if θ_M^R is an equivalence. Lemma 2.1 shows that good modules are closed under finite cofibers, shifts, and retracts. They are also closed under filtered colimits having a uniform upper homotopy bound. For such a diagram, the left side of (2.1) commutes with its colimit because relative tensor products do, and the right side commutes with it by Lemma 5.1. The forgetful functor from R -modules computes these colimits on underlying spectra.

First let G be a finitely generated abelian group. It admits a finite free resolution

$$0 \longrightarrow \mathbb{Z}^a \longrightarrow \mathbb{Z}^b \longrightarrow G \longrightarrow 0.$$

For completeness, the kernel of a surjection $\mathbb{Z}^b \rightarrow G$ is finite free: by induction on b , project a subgroup of $\mathbb{Z}^b$ onto the last coordinate. Its image is free of rank at most one, its kernel is a subgroup of $\mathbb{Z}^{b-1}$, and the resulting exact sequence splits. This proves the assertion about the kernel. The displayed resolution gives an R -linear cofiber sequence

$$(H\mathbb{Z})^a \longrightarrow (H\mathbb{Z})^b \longrightarrow HG.$$

Our hypothesis and finite exactness show that HG is good.

For an arbitrary abelian group G , its finitely generated subgroups form a filtered diagram with colimit G . Their Eilenberg–Mac Lane spectra all lie in degree zero and have colimit HG , as is checked on homotopy groups. Uniformly bounded-above continuity therefore makes HG good for every G .

Any R -module with homotopy in a finite interval is good by its finite Postnikov tower: its layers are shifts of $H(\pi_j M)$ in the heart of Mod_R . Finally, suppose $M \in (\operatorname{Mod}_R)_{\leq b}$ without any lower bound. There is a canonical equivalence

$$\operatorname{colim}_{r \geq 0} \tau_{\geq -r} M \simeq M. \quad (5.2)$$

It follows on homotopy groups, because each fixed homotopy group is unchanged once r is sufficiently large. Each term is zero or has homotopy in the finite interval $[-r, b]$ and is therefore good. All terms have the same upper bound b . The good-module closure just proved applies to (5.2), so M is good. $\square$

Corollary 5.3. *For every $n \geq 0$, every finite-type CW complex X , and every bounded-above $\tau_{\leq n} \mathbb{S}$ -module M , the natural map*

$$C^*(X, \tau_{\leq n} \mathbb{S}) \otimes_{\tau_{\leq n} \mathbb{S}} M \longrightarrow C^*(X, M)$$

is an equivalence.

Proof. Apply Proposition 5.2 to $R = \tau_{\leq n} \mathbb{S}$, using Theorem 4.2 with $P = \Sigma_+^\infty X$. $\square$

5.1. The base-change theorem.

Proof of Theorem 1.1. Choose $n \geq 0$ such that $\pi_i A = 0$ for $i > n$. The connective cover $A^c = \tau_{\geq 0} A$ has a commutative ring structure and a commutative ring map $A^c \rightarrow A$ [Lur, Proposition 7.1.3.13].

It is connective and n -truncated. The universal property of the ring Postnikov section of the unit map $\mathbb{S} \rightarrow A^c$ supplies commutative ring maps

$$\tau_{\leq n}\mathbb{S} \longrightarrow A^c \longrightarrow A \longrightarrow B; \quad (5.3)$$

see [Lur, Propositions 7.1.3.14–7.1.3.15]. Thus A, B are bounded-above $\tau_{\leq n}\mathbb{S}$ -modules. There is no requirement that their upper bounds agree.

Corollary 5.3 gives equivalences

$$\begin{aligned} \theta_A^{\tau_{\leq n}\mathbb{S}} : C^*(X, \tau_{\leq n}\mathbb{S}) \otimes_{\tau_{\leq n}\mathbb{S}} A &\xrightarrow{\sim} C^*(X, A), \\ \theta_B^{\tau_{\leq n}\mathbb{S}} : C^*(X, \tau_{\leq n}\mathbb{S}) \otimes_{\tau_{\leq n}\mathbb{S}} B &\xrightarrow{\sim} C^*(X, B). \end{aligned}$$

The first is right A -linear. To check this, use its defining evaluation map: multiplication by A on the coefficient factor commutes with evaluation, and the target is the function spectrum of an A -module. Hence it remains an equivalence after tensoring over A with B . Associativity of relative tensor products gives

$$\begin{aligned} C^*(X, A) \otimes_A B &\simeq (C^*(X, \tau_{\leq n}\mathbb{S}) \otimes_{\tau_{\leq n}\mathbb{S}} A) \otimes_A B \\ &\simeq C^*(X, \tau_{\leq n}\mathbb{S}) \otimes_{\tau_{\leq n}\mathbb{S}} B \\ &\xrightarrow{\theta_B^{\tau_{\leq n}\mathbb{S}}} C^*(X, B). \end{aligned}$$

This is the canonical comparison, not just an abstract equivalence of its source and target. More precisely, precomposing the canonical map (1.1) with $\theta_A^{\tau_{\leq n}\mathbb{S}} \otimes_A B$ gives $\theta_B^{\tau_{\leq n}\mathbb{S}}$ under the displayed associativity equivalence. Both composites are adjoint to evaluation followed by the coefficient map $\tau_{\leq n}\mathbb{S} \rightarrow A \rightarrow B$. The two already established equivalences therefore imply that (1.1) is an equivalence. $\square$

Remark 5.4 (Where the hypotheses are used). In the proof of Theorem 1.1, the boundedness of the sphere section gives a fixed width in Lemma 4.3. Finite homotopy groups make the Steenrod-module bounds and the finite ghost argument applicable. Finite type of X is used only in Lemma 5.1, and boundedness above of the eventual coefficients is used in the passage (5.2). No finite Tor-amplitude assertion about $A \rightarrow B$ is made or needed. In particular, the proof does not infer boundedness of a relative tensor product merely from boundedness of its two factors.

6. COUNTEREXAMPLES WITHOUT UPPER BOUNDS

The two examples in this section use rational polynomial ring spectra. We model these by commutative differential graded $\mathbb{Q}$ -algebras with zero differential, interpreting generator degrees homologically. Such algebras model E_∞ - $H\mathbb{Q}$ -algebras by [Lur, Proposition 7.1.4.11]; for the compatible algebraic description of module categories, see [Shi, Theorem 1.1 and Corollary 2.15]. Thus the polynomial symbols below specify commutative ring spectra, not merely graded homotopy rings. Neither example uses the bounded-above base-change theorem.

Lemma 6.1 (A common test space). *Let*

$$W = \bigvee_{n \geq 1} S^{2n},$$

with its usual CW topology and common basepoint. Then W is a simply connected finite-type CW complex, and for every spectrum M there is a natural equivalence

$$C^*(W, M) \simeq \prod_{n \geq 0} \Sigma^{-2n} M. \quad (6.1)$$

The $n = 0$ factor comes from the disjoint-basepoint suspension summand. For $M = H\mathbb{Q}$, the canonical map

$$\bigoplus_{n \geq 0} \Sigma^{-2n} H\mathbb{Q} \longrightarrow \prod_{n \geq 0} \Sigma^{-2n} H\mathbb{Q} \quad (6.2)$$

is an equivalence.

Proof. The space W has one 0-cell and one cell in each positive even dimension. Its 1-skeleton is a point, so cellular approximation gives simple connectivity. The basepoint and the collapse to a point split the suspension spectrum as

$$\Sigma_+^\infty W \simeq \mathbb{S} \oplus \bigoplus_{n \geq 1} \Sigma^{2n} \mathbb{S}.$$

Mapping out of a coproduct gives a product, proving (6.1). For a fixed integer j , at most one factor of (6.2) has nonzero π_j . Homotopy groups commute with products by representability. They also commute with direct sums: a direct sum is the filtered colimit of its finite subsums, and spheres are compact. Thus (6.2) induces an isomorphism on every homotopy group and is an equivalence. $\square$

6.1. Unbounded target.

Theorem 6.2. *Take $A = H\mathbb{Q}$, $B = H\mathbb{Q}[u]$ with $|u| = 2$, and $X = W$ as in Lemma 6.1. Then A is discrete and B is connective but not bounded above. The cochain base-change map is not an equivalence. More precisely, after identifying each $\mathbb{Q}u^n$ with $\mathbb{Q}$, its map on π_0 is*

$$\bigoplus_{n \geq 0} \mathbb{Q} \hookrightarrow \prod_{n \geq 0} \mathbb{Q}. \quad (6.3)$$

Proof. Lemma 6.1 and preservation of coproducts by tensor product give

$$C^*(W, A) \otimes_A B \simeq \left(\bigoplus_{n \geq 0} \Sigma^{-2n} A \right) \otimes_A B \simeq \bigoplus_{n \geq 0} \Sigma^{-2n} B.$$

On the other hand,

$$C^*(W, B) \simeq \prod_{n \geq 0} \Sigma^{-2n} B.$$

These identifications send the comparison to the canonical map from the coproduct to the product. Indeed, naturality with respect to the summands of $\Sigma_+^\infty W$ identifies its restriction to the n th summand with the identity in the n th coordinate and zero in every other coordinate. Since $\pi_{2n} B = \mathbb{Q}u^n$, taking π_0 gives (6.3). The sequence $(1, u, u^2, \dots)$ belongs to the target but not to the image. In particular, the cokernel is $\prod_{n \geq 0} \mathbb{Q} / \bigoplus_{n \geq 0} \mathbb{Q} \neq 0$. $\square$

Remark 6.3. A bounded-above source therefore does not suffice. Both coefficient rings in this example are connective, so connectivity does not repair the failure. The calculation involves no torsion, convergence issue, or stable-stem computation.

6.2. Unbounded source.

Theorem 6.4. *Let*

$$A = H\mathbb{Q}[x_1, x_2, x_3, \dots], \quad |x_j| = 2j, \quad B = H\mathbb{Q},$$

and let $A \rightarrow B$ be the augmentation sending every x_j to zero. For $X = W$, the cochain base-change map is not an equivalence. More precisely, there is an injective $\mathbb{Q}$ -linear map

$$\frac{\prod_{n \geq 1} \mathbb{Q}}{\bigoplus_{n \geq 1} \mathbb{Q}} \hookrightarrow \ker(\pi_0(C^*(W, A) \otimes_A B) \longrightarrow \pi_0 C^*(W, B)). \quad (6.4)$$

Here A is connective but unbounded above and B is discrete.

Proof. For $m \geq 0$, define the commutative A -algebra

$$A_m = H\mathbb{Q}[x_{m+1}, x_{m+2}, \dots],$$

where the structure map kills $x_1, \dots, x_m$, so $A_0 = A$. The transition $A_m \rightarrow A_{m+1}$ kills x_{m+1} . We will show that each A_m is perfect over A and that their filtered colimit is B . Lemma 2.1 will then identify the source of base change with $\operatorname{colim}_m C^*(W, A_m)$. In this colimit, a diagonal sequence of coefficients vanishes precisely when it has finite support; its image in $C^*(W, B)$ always vanishes. These two assertions give the injection in (6.4).

Finite perfect quotients. In the differential graded models, multiplication by x_{m+1} is injective and its cokernel is the next polynomial quotient. Consequently there is a cofiber sequence of A -modules

$$\Sigma^{2(m+1)} A_m \xrightarrow{x_{m+1}} A_m \longrightarrow A_{m+1}. \quad (6.5)$$

Indeed, the ordinary short exact sequence of these zero-differential complexes identifies the mapping cone of the first map with the quotient up to quasi-isomorphism. Passing to module spectra gives (6.5). Induction from $A_0 = A$ now proves that each A_m is perfect over A .

The augmentation as a filtered colimit. The natural map

$$\operatorname{colim}_m A_m \longrightarrow B \quad (6.6)$$

is an equivalence of A -modules. All terms are connective, the maps on π_0 are the identity of $\mathbb{Q}$, and for any fixed $d > 0$ the group $\pi_d A_m$ vanishes once $2(m+1) > d$. Homotopy groups commute with filtered colimits, proving the assertion.

The source of the comparison. By Lemma 2.1, perfectness of A_m gives natural equivalences

$$C^*(W, A) \otimes_A A_m \simeq C^*(W, A_m).$$

Since tensor product preserves colimits, (6.6) therefore gives

$$C^*(W, A) \otimes_A B \simeq \operatorname{colim}_m C^*(W, A_m). \quad (6.7)$$

Under this equivalence, naturality identifies the comparison to $C^*(W, B)$ with the map induced by the augmentations $A_m \rightarrow B$. This uses no interchange of the function-spectrum functor with the filtered colimit. It remains to find classes in this colimit that are nonzero but map to zero in $C^*(W, B)$.

Kernel classes. For $\lambda = (\lambda_n)_{n \geq 1} \in \prod_{n \geq 1} \mathbb{Q}$, consider

$$a(\lambda) = (0, \lambda_1 x_1, \lambda_2 x_2, \dots) \in \prod_{n \geq 0} \pi_{2n} A = \pi_0 C^*(W, A).$$

Its image at stage m of (6.7) has zero coordinates for $n \leq m$ and coordinate $\lambda_n x_n \in \pi_{2n} A_m$ for $n > m$. Since x_n is nonzero in A_m whenever $n > m$, this element is zero at stage m exactly when $\lambda_n = 0$ for every $n > m$. An element of a filtered colimit of abelian groups vanishes exactly when it vanishes at some subsequent stage. It follows that the kernel of the resulting map

$$\prod_{n \geq 1} \mathbb{Q} \longrightarrow \pi_0(C^*(W, A) \otimes_A B)$$

is precisely $\bigoplus_{n \geq 1} \mathbb{Q}$.

The image of every $a(\lambda)$ in $\pi_0 C^*(W, B)$ is zero: its $n = 0$ coordinate is zero by construction, and $\pi_{2n} B = 0$ for $n > 0$. Thus the induced injection from $\prod_{n \geq 1} \mathbb{Q} / \bigoplus_{n \geq 1} \mathbb{Q}$ lands in the kernel of base change, proving (6.4). Taking $\lambda_n = 1$ for every n gives an explicit nonzero kernel class. $\square$

Remark 6.5. In both examples, the coefficient rings have $\pi_0 = \mathbb{Q}$, no odd or negative homotopy groups, and finite-dimensional homotopy in each degree. For the source in Theorem 6.4, a degree $2d$ monomial corresponds to a partition of d , of which there are finitely many. Thus degreewise finite-dimensionality and connectivity do not replace the missing upper bound.

7. THE COMPARISON AFTER SKELETAL COMPLETION

Proposition 7.1. *Let X be a finite-type CW complex and $A \rightarrow B$ any map of commutative ring spectra. If $X^{(d)}$ denotes the d -skeleton, there is a natural equivalence in Mod_B*

$$\varprojlim_d (C^*(X^{(d)}, A) \otimes_A B) \simeq C^*(X, B). \quad (7.1)$$

Here the inverse limit is the limit in the stable ∞ -category Mod_B , hence a homotopy limit. The ordinary comparison factors through the natural map from $C^*(X, A) \otimes_A B$ to the inverse limit in (7.1).

Proof. Every $X^{(d)}$ is a finite CW complex. Corollary 2.3 therefore gives compatible equivalences

$$C^*(X^{(d)}, A) \otimes_A B \simeq C^*(X^{(d)}, B).$$

Moreover, $X \simeq \text{colim}_d X^{(d)}$, suspension spectra preserve this colimit, and mapping out converts it into a limit. Taking limits of the finite-skeleton comparisons proves (7.1). Naturality with respect to the inclusions $X^{(d)} \rightarrow X$ identifies their composite with the ordinary comparison and gives the stated factorization. $\square$

In Theorem 6.2, the finite-skeleton cochains are finite products of the shifts $\Sigma^{-2n}B$, and the restriction maps are the corresponding projections. Their limit is $\prod_{n \geq 0} \Sigma^{-2n}B$. Thus the map from the ordinary source to the source in (7.1) is visible on π_0 as the inclusion of finitely supported sequences into all sequences. In Theorem 6.4, the diagonal of independent positive-degree coefficients survives every finite quotient, although it maps to zero over the final discrete coefficient ring.

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